

Legacy Homogeneous Reactor Study

One-Group Diffusion and Retired Backends

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1 Purpose and status

This document records the first reactor-physics workflow developed for Reactor Kinetics Lab. It covers the homogeneous buckling estimate, the more detailed one-group annular r - z diffusion model, and the two backend services that were built from it.

These models are **legacy**. The frontend no longer exposes the one-group Core or Transient Diffusion pages. Current point kinetics uses an OpenMC continuous-energy rod-worth table, OpenMC-derived delayed-neutron data, and FMU thermal feedback. The current Core page uses resolved four-group diffusion. Those active models are documented in `reactor_model_theory.pdf`.

The legacy study remains useful for understanding the numerical development of the project and the motivation for moving to OpenMC. It also still supplies the 20 MW nominal operating scale, the displayed $1.5 \times 10^{12} \text{ n cm}^{-2} \text{ s}^{-1}$ flux normalization, and historical geometry metadata in `calibration.py`.

2 First diffusion model and transition to OpenMC

The original question was whether a heavy-water-moderated, natural-uranium reactor could be explored with a small, understandable diffusion model. The notebook `docs/oneGroupHomogeneous.ipynb` answered that question in two steps:

1. a homogeneous bare-cylinder buckling estimate for scale;
2. a piecewise one-group, two-dimensional annular diffusion solve for geometry, leakage, and rod motion.

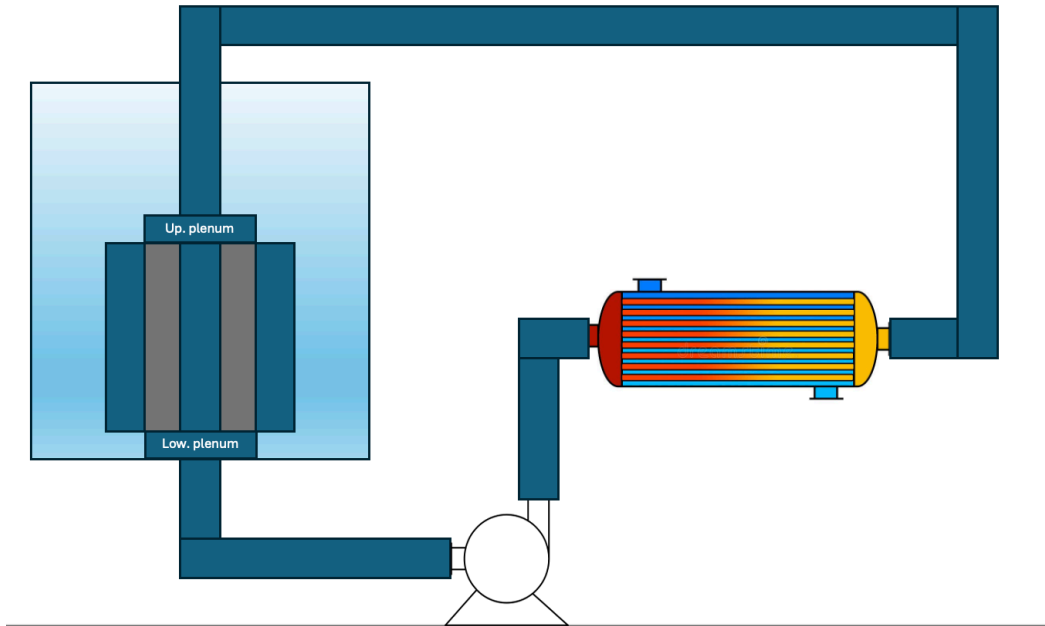


Figure 1: The exploratory homogeneous/annular reactor model used before the OpenMC reactor was developed.

The exercise established the sparse eigenvalue and transient methods later reused in the project. It also showed why the reactor definition could not remain a set of tuned one-group constants. The effective absorption had to represent unresolved spectral leakage, the fuel lattice was collapsed into one annulus, and the rod became an artificial absorbing cylinder. Because those choices were coupled, agreement in critical size could not independently validate material physics or rod worth. This limitation motivated the continuous-energy OpenMC model and the tracked reference-data workflow.

3 Estimate 1: homogeneous buckling

3.1 One-speed model

The first estimate uses one-speed diffusion theory in a homogeneous bare cylinder:

$$-D\nabla^2\phi + \Sigma_a\phi = \frac{1}{k}\nu\Sigma_f\phi.$$

For a critical homogeneous system, material and geometric buckling satisfy

$$B_m^2 = \frac{\nu\Sigma_f - \Sigma_a}{D} = B_g^2,$$

with the finite-cylinder approximation

$$B_g^2 \approx \left(\frac{2.405}{R}\right)^2 + \left(\frac{\pi}{H}\right)^2.$$

The estimate gives a first-pass critical size and checks the direction of leakage changes. It cannot represent the central moderator opening, separate reflector regions, axial reflector slabs, or a moving geometric rod.

3.2 Homogenization and power amplitude

Natural uranium and D₂O are volume-averaged into one effective core material. The criticality equation determines an eigenvalue and shape, not an absolute amplitude: if ϕ is a solution, then $C\phi$ is also a solution. The notebook therefore obtains a physical flux scale separately from

$$P = E_f \int_V \Sigma_f(\mathbf{r})\phi(\mathbf{r}) dV.$$

This distinction remains relevant in the current application. Point kinetics evolves a normalized neutron population and then applies independent nominal power and flux scales.

4 Estimate 2: one-group annular r - z model

4.1 Geometry

Estimate 2 solves a heterogeneous one-group diffusion eigenproblem on an axisymmetric cylindrical mesh:

$$-\nabla \cdot (D\nabla\phi) + \Sigma_a\phi = \frac{1}{k_{\text{eff}}}\nu\Sigma_f\phi.$$

The legacy frozen geometry is:

Quantity	Value
Inner moderator radius	80.0 cm
Fuel outer radius	345.6 cm
Reflector outer radius	405.6 cm
Active height	691.2 cm
Axial reflector thickness	60.0 cm per side
Equivalent rod radius	50.0 cm

Only $r \geq 0$ is meshed, but cylindrical volume weighting $2\pi r dr dz$ represents the complete three-dimensional body.

4.2 Engineering-effective constants

The frozen one-group values are:

Region	D [cm]	Σ_a [cm ⁻¹]	$\nu\Sigma_f$ [cm ⁻¹]
Inner D ₂ O moderator	1.25	3.5×10^{-4}	0
Fuel annulus	0.95	8.5×10^{-3}	8.59×10^{-3}
Outer D ₂ O reflector	1.15	4.0×10^{-4}	0

These are not transport-derived MGXS. They combine energy collapse, lattice homogenization, resonance and self-shielding effects, and unresolved leakage into one set of engineering-effective coefficients. The equivalent rod adds

$$\Delta\Sigma_{a,\text{rod,max}} = 0.25 \text{ cm}^{-1}$$

inside a central cylindrical region. It is a calibrated absorber, not a resolved control-rod material.

4.3 Boundary treatment and searches

The computational domain extends beyond the physical reflector by $2.13D_{\text{refl}}$ and imposes zero flux at that extrapolated boundary. The notebook uses the resulting solver in two outer calculations:

1. bisection on fuel radius until k_{eff} brackets unity;
2. repeated eigenvalue solves over rod insertion, with $\rho = (k - 1)/k$, to construct a diffusion-derived worth curve.

Warm-starting a rod solve from the preceding converged flux reduces iteration count without changing the physical model.

5 Retired steady Core backend

5.1 Spatial discretization

`diffusion.py` and `core_service.py` implement the notebook model on a cell-centered mesh with

$$\Delta r = \Delta z = 3 \text{ cm.}$$

Diffusion coefficients are harmonically averaged at material interfaces. Radial face terms include the cylindrical $r_{i\pm 1/2}$ weights, while axial terms use the usual center-to-center conductance.

After flattening the two-dimensional mesh, the eigenproblem is

$$L\phi = \frac{1}{k}F\phi,$$

where L is the sparse leakage-plus-absorption matrix and F is diagonal because fission production is local:

$$(F\phi)_{ij} = \nu\Sigma_{f,ij}\phi_{ij}.$$

5.2 Power iteration

The solver starts with a positive flux and repeatedly:

1. forms the fission source $F\phi$;
2. solves $L\phi^* = F\phi$ with a cached sparse factorization;
3. clips small negative roundoff values and peak-normalizes the flux;
4. updates k from the ratio of new and old total fission source.

The service returned a peak-normalized r - z field, radial and axial profiles, k_{eff} , and numerical metadata. Results were cached by rod insertion. The current frontend Core page no longer calls this service.

6 Retired direct transient-diffusion backend

6.1 Equations

`transient_diffusion.py` evolves a one-group spatial flux and six precursor fields:

$$\frac{1}{v} \frac{\partial \phi}{\partial t} = -L\phi + (1 - \beta) \frac{F}{k_{\text{ref}}} \phi + \sum_{i=1}^6 \lambda_i C_i, \quad (1)$$

$$\frac{\partial C_i}{\partial t} = \beta_i \frac{\nu\Sigma_f}{k_{\text{ref}}} \phi - \lambda_i C_i. \quad (2)$$

Here $v = 2.2 \times 10^5$ cm/s and k_{ref} is the initialization-state eigenvalue. Scaling the fission source by k_{ref} anchors the transient to the same discrete steady state.

6.2 Implicit time integration

Fully implicit Euler permits algebraic elimination of the precursor fields. One sparse flux system is solved per time step,

$$A\phi^{n+1} = b,$$

with

$$A = \frac{1}{v\Delta t} I + L - b_{\text{eff}} \text{diag} \left(\frac{\nu\Sigma_f}{k_{\text{ref}}} \right),$$

$$b_{\text{eff}} = (1 - \beta) + \sum_i \frac{\beta_i \lambda_i \Delta t}{1 + \lambda_i \Delta t}.$$

The precursor update is

$$C_i^{n+1} = \frac{C_i^n + \Delta t \beta_i (\nu \Sigma_f / k_{\text{ref}}) \phi^{n+1}}{1 + \lambda_i \Delta t}.$$

The retired service used a 5 cm spatial mesh and a 1 s step. It rebuilt and factorized the matrix when rod insertion changed, then reused that factorization between rod moves. The former frontend heatmap was downsampled to 40×60 points and history was limited to 200 samples.

6.3 Power normalization

Unlike point kinetics, the direct transient stores an amplitude-bearing spatial field. It therefore computes relative power from the cylindrical fission-rate integral:

$$\frac{P}{P_0} = \frac{\sum_j \nu \Sigma_{f,j} \phi_j V_j}{\sum_j \nu \Sigma_{f,j} \phi_{j,0} V_j}, \quad V_j \propto r_j \Delta r \Delta z.$$

The internal field is not peak-normalized after each step; only display copies are normalized. Otherwise physical growth and decay would be removed.

7 What remains in the current application

Most legacy calibration no longer drives active neutronics:

- point-kinetics rod worth comes from the OpenMC CE rod scan;
- delayed-neutron fractions and generation time come from the four-group OpenMC MGXS export;
- the Core page uses resolved four-group finite-volume diffusion.

Four legacy dependencies remain visible in current configuration:

- nominal power $P_0 = 20$ MW;
- displayed total-flux scale $\Phi_0 = 1.5 \times 10^{12} \text{ n cm}^{-2} \text{ s}^{-1}$;
- annular geometry metadata returned in the simulation model object; and
- the eight axial power fractions sent to the Modelica FMU, which are still computed by `core_service.compute_axial_power_fractions_8()`.

The flux scale should eventually be replaced with an absolute normalization derived consistently from the current OpenMC reactor. The FMU axial heat source should likewise be switched to the four-group CE-corrected power shape. These dependencies are retained here so their provenance remains clear rather than being mistaken for current OpenMC outputs.

8 Lessons and limitations

The one-group model was a successful numerical prototype but not a defensible current reactor definition. Its main limitations were:

- hand-selected energy collapse and effective loss coefficients;

- one homogenized fuel region instead of resolved fuel rings;
- an equivalent rod rather than transport-derived rod material data;
- fixed cross sections with no thermal-hydraulic feedback;
- no xenon, burnup, void, or detailed control-bank model.

The lasting contribution is methodological: it established cylindrical sparse operators, eigenvalue iteration, critical-size searches, rod scans, implicit precursor treatment, and the separation between normalized kinetics amplitude and physical display scaling. OpenMC then supplied the higher-fidelity reactor definition and reference data needed by the current application.